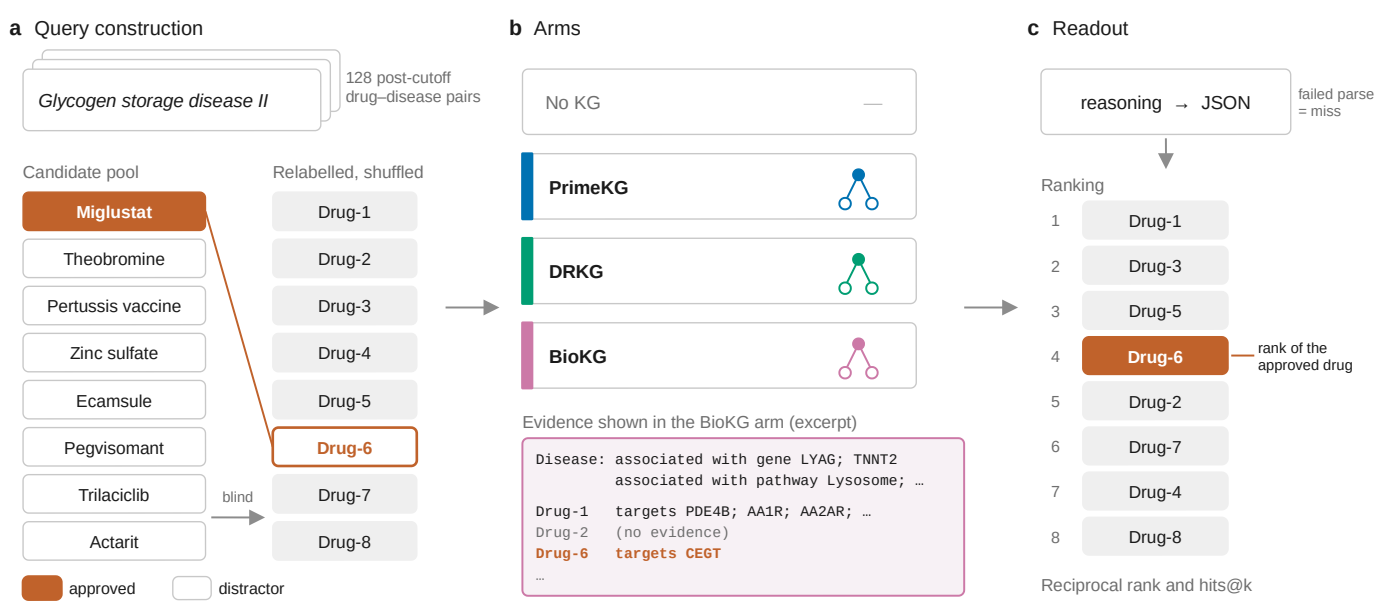

a Query construction
Glycogen storage disease II
128 post-cutoff drug–disease pairs
Candidate pool
Relabelled, shuffled
Miglustat
Theobromine
Pertussis vaccine
Zinc sulfate
Ecamsule
Pegvisomant
Trilaciclib
Actarit
blind
Drug-1
Drug-2
Drug-3
Drug-4
Drug-5
Drug-6
Drug-7
Drug-8
approved
distractor
b Arms
No KG
—
PrimeKG
DRKG
BioKG
Evidence shown in the BioKG arm (excerpt)
Disease: associated with gene LYAG; TNNT2
associated with pathway Lysosome; …
Drug-1 targets PDE4B; AA1R; AA2AR; …
Drug-2 (no evidence)
Drug-6 targets CEGT
…
c Readout
reasoning → JSON
failed parse = miss
Ranking
1 Drug-1
2 Drug-3
3 Drug-5
4 Drug-6
rank of the approved drug
5 Drug-2
6 Drug-7
7 Drug-4
8 Drug-8
Reciprocal rank and hits@k